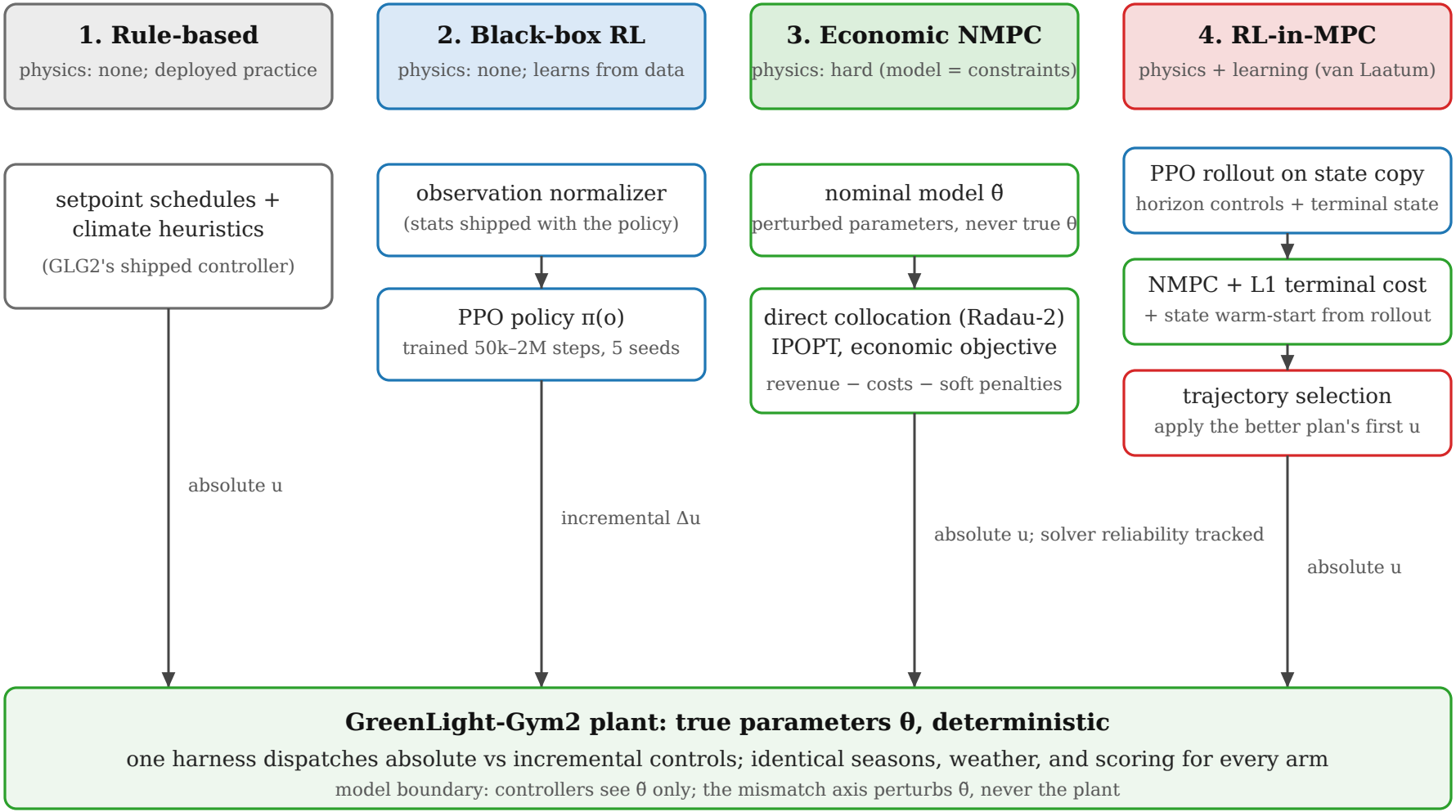Controller architectures: where physics enters, and how strongly